

Chapter 14

Microabrasive Technology for Precision Cleaning and Processing

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14.1 INTRODUCTION

Alternate surface cleaning methods are increasingly being sought and have been developed to replace conventional cleaning solvents, such as chlorinated compounds, hydrochlorofluorocarbons (HCFCs), trichloroethane, and other environmentally detrimental solvents. New regulations and mandates have been instituted for the reduction in the use of these solvents and their eventual phase-out.¹⁻³ Industrial manufacturers generally seek to minimize material losses, which typically requires very high precision and close tolerance processing operations such as removal of thin surface contaminant films and coatings, demarking components, drilling and cutting, etching, deburring, and surface texturing of a wide variety of materials from metals to plastics and ceramics. Macro- and micro-contamination from these operations is a significant detriment to achieving high production yields. Additionally, commercial industries need processing methods to better recover and recycle valuable materials, including materials from electronic circuit boards, dental waste, and chromium and gold plating from complex parts, while, at the same time, minimizing the waste generated and preventing pollution. These used items often contain other precious metals; hazardous materials, such as lead and cadmium solder; epoxy adhesives; radioactive sources; and PCB (polychlorinated biphenyls) compounds (e.g., capacitors). This combination of materials is designated as hazardous or mixed waste, and disposal of these items presents a major challenge in waste minimization.

Many alternative precision cleaning methods to solvent cleaning have been developed and several cleaning and processing methods are well established.⁴⁻⁷ Among these alternatives, microabrasive cleaning and processing is a versatile technology that employs a fine abrasive medium to impact the target surface to be cleaned and remove the contaminant material from the surface.⁸ The technique can be used to clean surfaces, remove coatings, or strip contaminants from a surface. It can also be used to remove loose material, blips, and burrs from production metal components after machining, and even soft materials such as plastics, polymers, and leather and rubber components can be processed.⁸⁻²¹ Other applications of the technology include component demarking, drilling and cutting, surface texturing, and etching a wide range of materials. Microabrasive technology can also address hazardous waste minimization needs since it can be used to recover precious metals, reduce waste generated during processing operations, and remove precisely radioactive and other hazardous contamination from selected areas of small complicated parts.

An overview of microabrasive precision cleaning and processing technology and its applications was published previously.⁸ Since that review there have been many developments of the technology and its applications. These recent developments are discussed in this chapter.

14.2 SURFACE CONTAMINATION AND SURFACE CLEANLINESS LEVELS

The most common categories of surface contaminants are given below.²² They can be present in a variety of states on the surface.

- Particles such as dust, metals, ceramics, glass, and plastics in the sub-micrometer to macro size range.
- Thin-film or molecular contamination that can be organic (hydrocarbons) or inorganic (acids gases and bases).
- Cationic (such as Na^+ and K^+) and anionic (such as Cl^- , F^- , SO_3^- , BO_3^{3-} , and PO_4^{2-}) contamination.
- Microbial contamination such as bacteria, fungi, and algae.

Other contaminant categories include metals, toxic and hazardous chemicals, radioactive materials, and biological substances on surfaces employed in specific industries, such as semiconductors, metals processing, chemical production, nuclear industry, pharmaceutical manufacture, and food processing, handling, and delivery. Common contamination sources include machining oils and greases, hydraulic and cleaning fluids, adhesives, waxes, human contamination, and particulates. In addition, a whole host of other chemical contaminants from a variety of sources may also soil a surface. Typical cleaning specifications are based on the amount of specific or characteristic contaminant remaining on the surface after it has been cleaned.

Cleanliness levels in precision technology applications are typically specified for particles by size (in the micrometer (μm) size range) and number of particles, as well as for hydrocarbon contamination represented by nonvolatile residue (NVR) in mass per unit area for surfaces or mass per unit volume for liquids.^{23–27} The cleanliness levels are based on contamination levels established in industry standard IEST-STD-CC1246D for particles from Level 1 to Level 1000 and for NVR from Level AA5 ($10 \text{ ng}/0.1 \text{ m}^2$) to Level J ($25 \text{ mg}/0.1 \text{ m}^2$).²⁶ The cleanliness levels have been revised or re-designated in revision E of this standard.²⁷ The maximum allowable number of particles for each particle size range has been rounded in revision E, while the NVR designation levels have been replaced with a single letter R followed by the maximum allowable mass of NVR. For example, former NVR Level J has the new designation R25; Level A/2 is now R5E-1; and Level AA5 is now R1E-5. These changes are discussed in greater detail in Volume 7 of this series.²²

Many of the products and manufacturing processes are also sensitive to, or they can even be destroyed by, airborne molecular contaminants (AMCs) that are present due to external, process or otherwise generated sources, making it essential to monitor and control AMCs. AMC is chemical contamination in the form of vapors or aerosols that can be organic or inorganic, and it includes everything from acids and bases to organometallic compounds and dopants.^{28,29} A new standard ISO 14644-10, “Cleanrooms and associated controlled

environments—Part 10: Classification of surface cleanliness by chemical concentration,³⁰ is now available as an international standard that defines the classification system for cleanliness of surfaces in cleanrooms with respect to the presence of chemical compounds or elements (including molecules, ions, atoms, and particles).

In many commercial applications, the precision cleanliness level is defined as an organic contaminant level of less than 10 µg of contaminant per cm², although for many applications the requirement is set at 1 µg/cm².^{26,27} These cleanliness levels are either very desirable or are required by the function of parts such as medical devices, electronic assemblies, optical and laser components, precision mechanical parts, and computer parts.

14.3 FUNDAMENTAL CONSIDERATIONS

Microabrasive precision cleaning involves removal of two types of surface contaminants: thin films or coatings, and particles. The mechanisms of removal of each of these contaminants are different and are discussed below.

14.3.1 Removal of Surface Films and Coatings

Many fundamental and experimental investigations have been carried out to understand material removal mechanisms during the particle impact process, including molecular dynamics, Monte Carlo and finite element numerical simulation of single and multiple particle impact.^{31–87} Dynamic effects, such as wave reflection, are negligible and the impact can be treated as quasi-static. Coatings with high interfacial strength are removed by mechanical erosion, while low interfacial strength results in coating removal by delamination.

The work necessary to remove a film is supplied by the kinetic energy E_{KE} of the particle as it impacts the surface:

$$E_{KE} = \frac{1}{2}mv^2 \quad (14.1)$$

Here m is the mass of an individual particle and v is the initial velocity.

The erosion rate is proportional to the ratio of the energy required for plastic deformation and the energy required for subsequent fracture, and is, in turn, dependent on the properties of the impacting particles and the target material (hardness, density, Poisson coefficients, elastic modulus, toughness, density), and the angle of impact.

14.3.1.1 Removal by Mechanical Erosion

In the case of removal of ductile coatings with high interfacial strength, three basic mechanisms of mechanical erosion have been identified: plowing erosion and two types of cutting erosion. Plowing deformation is usually caused by spherical particles or by the rounded surface of an irregular particle impacting the

surface. Material is displaced from the impact crater, part of which forms a lip at the exit end of the crater and part of which may form ridges around the sides of the crater. This material at the lip can be removed by subsequent impacts.

In the case of angular particles, the orientation of the particle at impact also becomes important. Depending on the angle of orientation (-17° to -90° , at a 30° impact angle), the particle rebounds from the surface with appreciable rotational velocity in the forward direction. Under these plane-strain conditions, all the displaced material is extruded to a large lip at the crater exit, and is removed by subsequent secondary impacts of the cartwheeling particle over the surface. This is referred to as Type I cutting action.

For orientation angles between 0° and -17° at a 30° impact angle, the particle rotates backwards during impact and all material is removed from the impact due to the machining action of the rotating particle. This is referred to Type II cutting action. Since Type II cutting occurs over a narrower range of orientation angles, it would be observed in only about one-sixth of the impacts of randomly oriented particles. This explains the lack of experimental evidence for a machining mechanism in erosion.

Neglecting elastic effects, the kinematics of the particle traveling through the semi-infinite metal substrate can be modeled as being resisted by a force given by

$$F_{\text{Resis}} = CP_{\text{Flow}}A_C \quad (14.2)$$

where F_{Resis} is the resistance force, C is a constant, P_{Flow} is the plastic flow pressure, and A_C is the contact area. The plastic flow pressure is the contact pressure value reached when the coating has fully yielded.

The main drawback of this approach is that the entire energy of the particle impact has to be consumed in plastic deformation, which leads to a prediction of zero rebound velocity and springback. In realistic collisions, this is clearly not the case, since there will be components of normal rebound velocity and springback in particle impacts at any angle of incidence.

If elastic effects are included, a realistic elastic-plastic analysis of particle impact can be developed.^{60,61} In this case, the particle first encounters an elastic retarding force (Figure 14.1). The particle then experiences an elastic-plastic transition retarding force, which is assumed to be the incident elastic force until fully plastic conditions are reached. The plastic retarding force is in effect until the point of maximum penetration, where the particle velocity in the y -direction is zero. Finally, the particle encounters an elastic rebound contact force, resulting in acceleration away from the surface. This rebound force is assumed to act in the y -direction.

From this analysis, the rebound velocity, the rebound angle, and the crater size and shape can be calculated and are found to be in excellent agreement (within 1%) with available experimental data.^{62,65} Accurate predictions of crater depth are important in microabrasive cleaning applications, since the material displaced to the crater lip is removed by subsequent impacts.

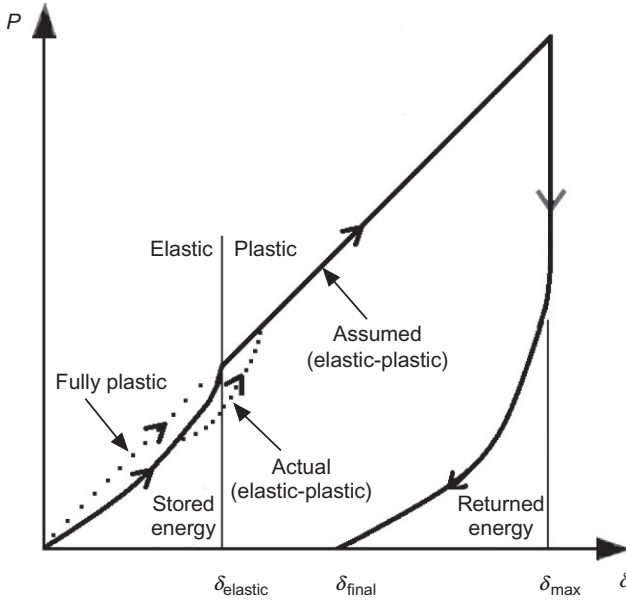


FIGURE 14.1 Force-deflection curve for fully plastic and elastic-plastic particle impact.⁴⁶

14.3.1.2 Removal by Delamination

Removal of a coating by delamination is a three-stage process. In the first stage, delamination is initiated on particle impact due to the large shear stress at the interface. However, the shear stresses are not responsible for removal of the coating since their effect is highly localized. The stresses become negligible at a distance of 0.6–0.8 mm from the point of impact.

On impact, the particle causes an indentation of the coating. This penetration induces compressive stresses in the coating. Together with the initial delaminated region, the stresses can cause buckling of the coating (stage 2) and crack initiation when a critical stress value is reached.^{36–38} Delamination propagates until the particle rebounds from the surface (stage 3). A delamination criterion has been proposed as:

$$\left(\frac{\langle \sigma_N \rangle}{\sigma_{NF}} \right)^2 + \left(\frac{|\sigma_S|}{\sigma_{SF}} \right) \leq 1 \quad (14.3)$$

where $\langle \sigma_N \rangle$ is the positive part of the stress vector normal to the interface, σ_S is the shear stress, σ_{NF} is the normal failure stress, and σ_{SF} is the shear failure stress. Failure is assumed to occur if the left-hand side of Equation (14.3) is greater than 1.

Material removal is observed even on the atomic scale. Wear is due to removal and rearrangement of individual metal atoms or single ion pairs, for metallic substrates and ionic crystals respectively, rather than plastic deformation and fracture, and can be modeled by stress-assisted chemical reaction kinetics.^{88–97}

14.3.2 Removal of Particles

Particles smaller than 10 μm adhere very strongly to the surface. For a dry surface, the dominant adhesion force is the van der Waals force [Ref. 98 and other references cited therein]. This force of adhesion can also be treated as the force necessary to remove the particle from the surface. For a 0.5- μm alumina particle on a silicon substrate, the adhesion force is estimated to be 3×10^{-9} N.⁹⁸ The estimated impact force of a 50- μm alumina particle in a microabrasive system operating at 0.6 MPa is 2×10^{-3} N, which is significantly greater than the adhesion force of the particle, thus making it possible to easily remove a 0.5- μm alumina particle.

14.4 MICROABRASIVE TECHNOLOGY

Microabrasive technology is a versatile, dry cleaning and processing technique that has wide application in precision cleaning and processing of a wide variety of materials.^{8–21} Since the initial U.S. patent for abrasive cleaning,⁹⁹ microabrasive technology has been developed to a very high degree of sophistication for a wide range of precision cleaning and processing applications.^{8,100–149} Many theoretical and experimental studies have been carried out to understand the key parameters that control the process.^{150–167}

14.4.1 Description of the Process

The microabrasive process uses a miniature nozzle through which a precisely graded abrasive powder, mixed with compressed air or other inert gas, is directed onto the surface of the part to be cleaned. The process is controlled by several key parameters:

- *Air or other gas pressure.* Gas pressure directly affects the strength of the abrasive stream and, in turn, the material removed. Higher pressure produces a stronger abrasive mixture and greater abrasive action.
- *Abrasive flow rate.* This is related to the gas pressure. At higher abrasive flow rates, more precise control can be applied to removal of material from the surface.
- *Nozzle size and shape.* The size of the nozzle determines the surface area that is impacted by the abrasive. The nozzle shape affects the ability to access tight confined spaces or wide area coverage of the surface of the part.

- *Nozzle distance from the surface of the part.* The distance of the nozzle from the surface determines the rate and precision of material removal.
- *Angle of impingement.* The angle of impingement determines material removal rate in ductile (incident angle between 15° and 45°) and brittle (incident angle 90° for maximum removal rate) materials.
- *Type and grade of abrasive.* The size, shape, and hardness of the abrasive determines the effectiveness of the process.
- *Dwell time.* Longer dwell times generally result in greater material removal rates.

The velocity V_{Exit} of a particle exiting from the nozzle is given by¹⁶⁸:

$$V_{\text{Exit}} = V_{\text{Air}} - \frac{1}{\alpha \Delta t + 1/(V_{\text{Air}} - V_o)} \quad (14.4)$$

$$\alpha = \frac{3C_D \rho_{\text{Air}}}{4\rho_{\text{Part}} d_{\text{Part}}} \quad (14.5)$$

$$X_L = V_{\text{Air}} \Delta t - \frac{1}{\alpha} \ln[\alpha(V_{\text{Air}} - V_o) + 1] \quad (14.6)$$

where V_{Exit} is the exit velocity of a particle; V_{Air} is the air velocity in the nozzle; C_D is the drag coefficient; Δt is the duration of particle acceleration in the nozzle; V_o is the initial velocity of the particle; ρ_{Air} is the density of air; ρ_{Part} is the density of the particle; d_{Part} is the particle diameter; and X_L is the length of the nozzle throat.

For an alumina particle of 100 μm diameter accelerated through a nozzle at 0.6 MPa, the exit velocity of the particle is over 170 m/s. The kinetic energy of the particle decreases with decreasing particle diameter. A 500- μm particle has nearly seven times the kinetic energy of a 100- μm particle. However, the total kinetic energy of a given mass of particle increases as the average particle diameter decreases.

The shape, size, and hardness of the particles are also important in cleaning. Abrasive media with smaller average particle size are generally more aggressive for given cleaning conditions due also, in part, to the more uniform shape of the particles.

Particles with high hardness (>7 on the Mohs hardness scale) remain intact on impact and the force is directed largely into penetrating the substrate. At the same time, the particles tend to bounce off the surface with little or no material removal. By contrast, softer particles tend to shatter on impact and part of the kinetic energy is directed along the surface. Some of the particle fragments travel along the surface, significantly increasing the removal efficiency of the contaminant film.

During impact the particle is subjected to very high stresses that will cause the particle to deform. The substrate also consumes a large amount of energy which consists of two components, kinetic energy and internal energy. The substrate undergoes greater plastic deformation and damage in normal particle

impact (90°) than in oblique impact ($<60^\circ$). This is supported by the observation that the kinetic energy of particle rebound is greater in oblique impact than in normal impact.⁵⁴

14.4.2 System Description

A basic microabrasive system includes an abrasive mixer, the work chamber, an air dryer and filter, a dust collector, and the nozzles and the abrasive media (Figure 14.2). The abrasive is mixed with filtered dry air (or other inert gas for reactive materials) in a mixer that independently controls the air pressure (typically 0.1–1.2 MPa) and powder flow rate (typically 0.2–1 kg/h). This permits very gentle or very hard processing and reduces the amount of powder necessary for each operation, thus minimizing the volume of secondary waste generated.

The system incorporates a drying unit because dry and filtered air or other gas is required to ensure appropriate operation of the abrasive mixer. The abrasive mixer has adjustable air pressure and independent power flow regulation to allow proper mixing of the powder and to permit optimum process control.

A vacuum shield around the nozzle (Figure 14.2) directly exhausts the powder and contaminants removed from the part. This shield confines the abrasive powder and the removed contaminants to a local area to prevent cross-contamination. The shield is contained within a high-efficiency particulate air (HEPA)-filtered work chamber and is itself connected through a HEPA

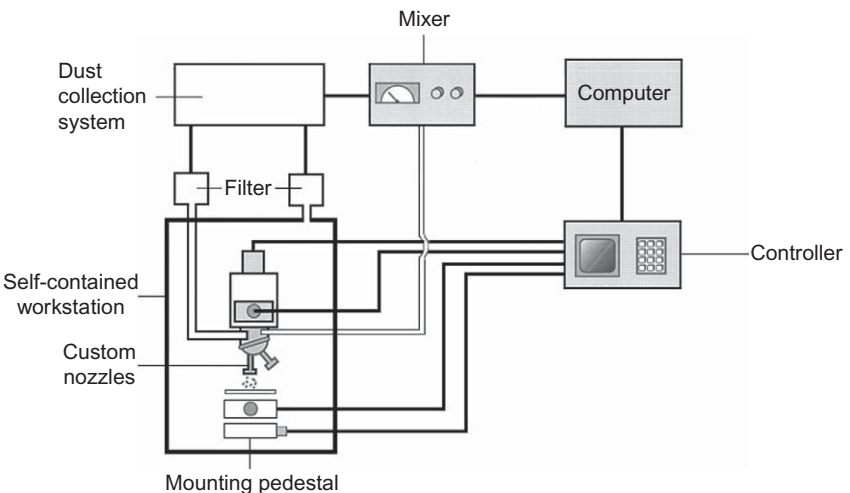


FIGURE 14.2 Typical microabrasive system for precision cleaning and processing applications.⁸

filter to a dust collector. The HEPA filters and the dust collection unit contain the contaminants and the abrasive powder, thereby preventing external pollution.

All processing operations are performed within an enclosed work chamber. For high-precision repeatable operation, the work chamber includes an automated pedestal on which parts are mounted for processing. The part to be cleaned is mounted on a computer-controlled, multi-axis positioner that allows automated precision processing operations. The computer control system is used to manipulate the multi-axis positioner for the mounting pedestal and to control the abrasive mixing unit. User-friendly software enables the operator to perform the various process functions. These functions are inputted into the controller, which signals the X-Y-Z and rotation positioners to initiate operations. An automated system can increase productivity for each application by allowing the operator to choose coordinates and speeds that maximize machine usage and minimize the amount of abrasive powder used.

14.4.3 Nozzle Materials and Design

There is a wide range of nozzles available from commercial suppliers^{169–184} for different applications, including right-angle and oblique-angle designs with extended lengths that make it possible to access tight spaces. An appropriate nozzle makes it possible to perform selective removal of materials without affecting the surrounding surface. Nozzle design selection is dependent on the powder used and the intended operation. For example, wide fan-shaped nozzles are used for overall cleaning or preparation of wide areas, including etching and frosting of glass surfaces; round and angular nozzles are suitable for boring holes or for cutting and removing circuit board components.

The smallest commercially available nozzles have 100- μm openings. In order to extend the range of the technology to finer applications, new patented techniques have been developed to fabricate nozzles with openings as small as 20 μm .⁸ These ultrafine nozzles can be fabricated from stainless steels, ceramics such as boron nitride or silicon carbide, or refractory metals such as molybdenum.⁸ This now makes it possible to use the system for precision micro-fabrication of refractory materials. The different abrasives mentioned below have been used with nozzles that have openings ranging from 20 to 150 μm .

New high-performance nozzles are available with a tighter focus and higher particle velocity. This reduces the overspray and processing times for applications such as cutting and deburring. To increase productivity for manufacturing operations, different multi-nozzle assemblies have been designed. These nozzle assemblies incorporate the custom-designed vacuum shields that are customized to the nozzle assembly design. The multi-nozzle assemblies with multiple axis control enable accurate and precise processing of complex shapes and parts with tight spaces.

14.4.4 Types of Abrasives

Commonly used abrasives include silicon carbide, alumina, sodium bicarbonate, dolomite (calcium carbonate), and softer media such as walnut shell, plastic and glass beads. Table 14.1 lists the relevant properties of the abrasives and typical applications for which these abrasives are used.

Silicon carbide is very aggressive and suitable for boring holes and cutting and removing electronic circuit board components, as well as for deburring metal components. Alumina is less aggressive and is suitable for precise surface cleaning and for removing component identification labels, such as those on hand tools. It is also suitable for etching glass or for cutting precise grooves in thin-film coatings on glass panels. Sodium bicarbonate and dolomite are excellent for wide-area surface cleaning, while softer biodegradable materials, such as walnut shell and plastic bead, are suitable for use on surfaces of critical electronic or optical components. Newer abrasives that have been successfully tested include diamond, cubic boron nitride, ceria, titanium dioxide, and tungsten carbide, which are ideal for processing very hard materials, such as refractory metals and alloys and ceramics.⁸ Recently, environmentally friendly water-soluble abrasives, such as kieserite and wheat or corn starch, have been introduced for precision cleaning and processing operations.^{172,185} The abrasives can be washed off completely after use. With a Mohs hardness of 3.5, kieserite is suitable for deburring medical-grade metals, removing metallic coatings and organic contaminants from non-metallic components, and fossil and mineral preparation. In a novel application, industrial wastes such as coal slag and smelter slag have been utilized as microabrasives for removing contaminants from metal surfaces.¹⁸⁶

14.4.4.1 Selection of the Abrasive

The selection of a particular abrasive is determined by its action for the type of function to be performed on the material to be processed. The effect of an abrasive material depends on three key characteristics:

- a. *Particle shape.* The shape of an abrasive may be angular, blocky, semi-round or spherical. Angular particles cut and strip away surface material on impact. Spherical particles do not have any cutting edges and are used to pound or peen a surface. Grain shape is important as rounded and angular grains behave differently when they impact a substrate such as steel. Angular, sharp particles produce the greatest cutting action and the deepest profile. Round or semi-round particles will cut much more slowly and will produce a more shallow profile.
- b. *Hardness.* The hardness of an abrasive is measured using the Mohs scale. Harder particles will be more aggressive in removing surface material.
- c. *Particle size.* A larger particle removes material faster and it tends to produce a heavier texture or rougher surface on the base material.

TABLE 14.1 Abrasive Media for Microabrasive Cleaning and Processing Applications⁸

Abrasive	Hardness (Mohs scale)	Average Particle Size (μm)	Description/Characteristics	Applications
Diamond	10	0.25-5	Diamond has the highest hardness known. The small particle size makes it possible to achieve high-quality surface finish and high-precision micromachining	Processing of hard metals and ceramics
Cubic boron nitride (CBN)	9.5	1-10	CBN has the second highest hardness known. The small particle size makes it possible to achieve high-quality surface finish and high-precision micromachining	Processing of hard metals and ceramics
Silicon carbide (SiC)	9.3	20-60	This is the most aggressive powder used for microabrasive blasting. It is typically used when very fast material removal is a requirement	Cutting, deburring, drilling, texturing, fossil preparation, wafer fabrication
Aluminum oxide (Al ₂ O ₃)	9	12-165	Alumina is the most commonly used cutting abrasive. The shape and hardness of the particles make it an excellent choice when working with metals or hard, brittle parts	Intricate cutting; hole and slot drilling; surface preparation; fine surface finishes; heavy deburring; large-scale rust removal; wafer fabrication; surface texturing; fossil preparation
Crushed glass	5.5	50-60	Crushed glass results in a material that works well as a mild abrasive. It has the hardness of glass bead with numerous shard-like edges. Crushed glass is most often used when only a light degree of abrasion is required	Light cleaning; surface finishing; epoxy removal; mineral preparation; restoration of artifacts
Walnut shell	4	200-250	The hardness of walnut shell falls between sodium bicarbonate and glass bead	It will quickly remove polymer coatings from circuit board surfaces, clean metal or ceramic surfaces without altering the surface finish, and deflash plastic parts

Plastic bead	3.5	150-200	The large size of plastic bead makes it an effective medium to deburr machined plastic parts without causing dimensional changes. It is good for stripping soft materials, such as paint or conformal coatings from hard substrates	Deflashing plastic component lead frames without removing the plating; deburring Delrin®, nylon and Teflon®; conformal coating removal; fossil preparation
Glass bead	5.5	40-50	This medium is commonly used for the critical preservation of tight tolerances combined with the need to relieve machining stresses, light deburring, or putting a satin-like finish on a part. The spherical shape of the bead keeps it from cutting into the surface of the part, so it is often used to relieve stresses by “peening” the surface of the part	Mild cleaning; light deburring; light surface finishing; peening; polishing; mineral preparation
Sodium bicarbonate	2.5	50-75	Sodium bicarbonate is one of the softest abrasives used. Combined with the “needle-like” shape of the particles, it is an excellent choice for abrading more pliable materials. The particles tend to cut through softer surfaces. It is commonly used to selectively remove the coating on a circuit board without damaging the individual components. Sodium bicarbonate is water soluble, making it easy to remove from delicate parts and making it highly amenable to waste management	Conformal coating removal; cleaning; demarking of plastic and gold lid devices; solder mask removal; deburring Delrin®, nylon and Teflon®; fossil preparation
Dolomite (calcium carbonate)	3.5	40-50	The characteristics are similar to sodium bicarbonate	Light cleaning; epoxy removal; fossil preparation

14.4.4.2 Abrasive Quality

The microabrasive process is strongly affected by the quality of the abrasive. There are three main quality issues that cause the most concern: moisture, foreign particles, and poor particle size distribution.

- a. *Moisture.* The average particle size of a microabrasive powder ranges from 0.25 to 200 μm . The small size of the particles increases the surface area relative to volume and moisture is readily absorbed from the ambient air and the compressed air source. As the particles absorb moisture they tend to clump together. These clumps cause uneven flow and even total blockage.
- b. *Foreign particles.* The small size of the orifices and nozzles used in microabrasive technology make the process sensitive to contaminants that would pass through larger standard blasting machines. Contaminants such as fibers, plastic, and even wood must be removed to produce an effective microabrasive powder.
- c. *Particle size distribution.* In a high-quality powder, a majority of the particles are located within a narrow size range and only a few particles are found at the coarse and fine extremes. The large particles are more likely to plug the orifice and the nozzle, and they will have a more aggressive effect on the part being processed. The effects of very fine particles can be equally damaging to the process. Fine particles will tend to be more sensitive to moisture. After being poured into a tank, the small particles will fill in the air spaces between the larger particles. As the air spaces are removed the abrasive particles will pack together more tightly, increasing the amount of energy required to make the medium flow effectively or even stopping the flow completely.

To maintain high quality, the abrasive powder is carefully processed in a humidity- and temperature-controlled environment. The moisture content and size distribution are carefully monitored to ensure that the abrasive meets stringent standards. The result is an abrasive powder that performs efficiently and consistently from lot to lot.

14.4.5 Air Treatment

The presence of even a small amount of moisture can cause the powder to clump and prevent it from flowing freely from the tank, bringing the entire operation to an abrupt halt.

Atmospheric air always contains moisture. It becomes a problem, even in arid regions, because of the nature of the compression process. Atmospheric air compressed to 0.6 MPa is one-eighth its original volume yet still contains the same amount of moisture. Additionally, the air gets hot as it is compressed, then cools as it leaves the compressor and travels throughout the system. Any significant drop in temperature or increase in pressure will cause moisture to condense out of the air.

To prevent moisture contamination, it is essential that the powder is stored in tightly closed containers in a cool, dry location when not in use. In addition, the power should not be left in the machine mixing tank for extended times when the machine is turned off and is depressurized. For the most efficient operation of the system, clean and dry air is required. This can be achieved by installing an adequate air treatment system to remove water, oil, and other particulate matter.

14.4.5.1 Air Drying

Drying equipment is rated on the dew point, which defines how cold the air supply can get before water begins to condense. The greater the volume of moisture in the compressed air line, the higher the dew point. Some types of abrasive media may be more or less susceptible to moisture than others, but the microabrasive blasting process requires that the moisture content in the air be below 200 parts per million (ppm). This corresponds to a dew point of 241 K (-32°C).

Most microabrasive operations incorporate a point-of-use air dryer with the blasting unit. This is a cost-effective solution, since the air dryer only needs to be sized according to the machine needs.

Several types of compressed air treatment systems are available which may be used instead of, or in conjunction with, a refrigerant dryer, including desiccant and membrane air dryers.

- a. Desiccant dryers use a super-absorbent bead that extracts moisture out of the compressed air as it passes through the dryer. When the desiccant beads are fresh they can achieve dew points below 236 K (-36°C). Most manufacturers use a bead that changes color when the bead becomes progressively saturated, typically blue to clear. When the beads become clear they will need to be either regenerated or replaced in steady use. Depending on the relative humidity, this may be required as often as once a week. To regenerate the desiccant, the moisture is removed by heating the beads. Each recharge causes the beads to lose some of their absorbing capacity. This significant amount of maintenance normally limits the use of a desiccant dryer to intermittent applications where the amount of air being dried is minimal and low cost is critical.
- b. Membrane dryers use semipermeable hollow membranes of fibers in which water vapor will diffuse through the fiber walls (osmosis). The moisture is removed from the unit through an automatic air purge. Membrane dryers can achieve dew points as low as 233 K (-40°C). The air purge continuously removes the excess water vapor from the system, eliminating the maintenance requirement associated with desiccant dryers. Typically the initial cost for a membrane dryer is higher, but in heavy-use environments it is more than offset by the lower maintenance cost.

An integrated dryer/filter consisting of the following three stages is also available:

- First stage: Separation. Entering air is spun around a stationary cone, coalescing water and oil droplets, breaking up brittle scale, and discharging the moisture through the drain mechanism.
- Second stage: Drying. A desiccant, contained in a non-absorbent bag, draws water and oil vapors out of the air flow. Medium-sized particles are prohibited from passage.
- Third stage: Filtration. A multilayer disk pack of final filters removes particles to 1 μm in size.

14.4.6 Oil Contamination

Initial oil contamination will cause irregularities in the abrasive flow. Clumps in the powder can be formed above the orifice in the abrasive tank, which limits the amount of abrasive that can pass through. Alternately, the clumps will pass through the orifice and obstruct the nozzle.

The major impact of oil contamination is check valve failure, which will prevent the check valve from sealing properly. When this happens, the air carrying some abrasive is allowed to pass through the check valve into the clean side of the system. The aggressive nature of the abrasive will quickly erode the O-rings and seals of the valves and cylinders in the system, nearly always requiring replacement of the system.

Microabrasive processing requires the oil in the air source be no more than 10 ppm. Since oil molecules are relatively large, this is easily and inexpensively accomplished with a simple in-line coalescing filter. Additionally, the oil will bond with the agents used to dry the compressed air, limiting their effectiveness. As the dryer becomes coated with oil it will not absorb moisture, allowing it to pass through to the machine. Therefore, the oil filter must be installed upstream from the air dryer.

14.4.7 Dust Collection

Industrial dust collectors provide high-efficiency air cleaning. With the fan on the clean air side, dust and other collected material are filtered out by the fabric filter before reaching the fan, resulting in long, reliable performance. The envelope filters provide very high collection efficiencies, trapping dust particles of all sizes, including those smaller than 1 μm . They are rated at 99.9% collection efficiency by weight even with high concentrations of sub-micrometer particles in the air stream. By adding a HEPA filter, the rating is 99.97% efficiency at 0.3 μm .

14.4.8 Recycling and Secondary Waste

Microabrasive processing is unique in that it cycles the abrasive medium through the system only once. Used media cannot be reclaimed and reprocessed into material that is suitable for use a second time for the following reasons:

- The blasting process causes the abrasive to break down very quickly, which changes the particle size distribution and causes the cutting force of the particles to change. Process reliability cannot be assured.
- When the abrasive is exposed to air it will readily absorb moisture, degrading the process as discussed earlier.
- Fragments of the substrate get mixed in with the abrasive, resulting in contamination and changed abrasive characteristic of the medium.

The use of dry abrasive powders eliminates the need for aqueous cleaning agents, which are difficult and expensive to process and dispose of. None of the available abrasives generates additional secondary waste. Sodium bicarbonate and dolomite are also water soluble, making waste treatment and handling easier. The expensive abrasives, such as diamond and boron nitride, can be recovered from the waste stream with optional patented separation systems for use in other applications. Depending on the abrasive and the waste stream, different aqueous or non-aqueous waste separation methods, such as electrostatic precipitation, electroacoustic separation or flocculation, are used to separate and recover the abrasives.¹⁸⁷

14.4.9 Static Charging

A novel point-of-use ionizer is available which effectively reduces static charge from reaching levels that could potentially cause damage to electrostatic discharge (ESD)-sensitive components. The point ionizer introduces a balanced flow of positive and negative ions into the abrasive pathway at the exit point of the abrasive nozzle tip. The abrasive powder maintains a much lower charge upon contact with the work piece, essentially eliminating the potential for ESD damage. The hand piece combines the abrasive nozzle and the ionization sources. A simple grounding point for the nozzle within the hand piece further neutralizes the abrasive pathway. The nozzle is positioned at a fixed angle to the ionization source.

The effectiveness of the point ionizer has been demonstrated during several extensive studies of conformal coating removal. Using different abrasive media with the point ionizer, the conformal coating was selectively removed while safely maintaining surface voltages below 20 V. Without the use of the point ionizer, charges of 3000 V and higher were measured.¹⁷²

In addition to reducing the risk of ESD damage, the point ionizer also improves the cleanliness on surfaces by preventing particles from adhering to the surface of the work piece being blasted due to the electrical force.⁸

14.5 CLEANING AND PROCESSING SYSTEMS

A wide range of microabrasive systems is commercially available, ranging from small, manually operated units to large integrated industrial units.^{169–184} Figure 14.3 shows examples of several systems that can be used for microabrasive cleaning and processing. Parts are processed in a self-contained enclosed work chamber that can be equipped with various accessories, such as ultraviolet and white lights and microscopes, for close and precision work. These units are



FIGURE 14.3 (a) Small benchtop microblaster unit.¹⁷² (b) Single- and dual-tank microblaster units.^{171,172} (c) Large-capacity microblaster units.¹⁷⁰ (d) Cabinet-style microblaster units with multiple abrasive tanks.^{183,184} (e) Large-capacity system for cleaning and restoration of facades and monuments.¹⁷⁹ (f) Stand-alone work cell with abrasive blaster.¹⁷² (Panel (a): Courtesy of Crystal Mark, Inc. Panel (b): (i) Courtesy of Crystal Mark, Inc. and (ii) Courtesy of Comco, Inc. Panel (c): Courtesy of S. S. White Technologies, Inc. Panel (d): (i) Courtesy of Vaniman Manufacturing Company and (ii) Courtesy of Texas Airsonics, Inc. Panel (e): Courtesy of Quintek Corporation. Panel (f): Courtesy of Crystal Mark, Inc.)

often equipped with multiple abrasive storage tanks that allow the operator to quickly switch from one abrasive to a different abrasive for various processing operations, or to perform continuous extended operations without interruption to refill the abrasive tank. Commercial suppliers also offer different abrasive powders that can be used with the nozzles appropriate for the part to be processed. Abrasive consumption is typically 0.2-1 kg/h, depending on the operation.

The microabrasive systems can be scaled to large machines designed to operate continuously under high-production environments. An example of such a system is shown in [Figure 14.4](#). The mobilized production unit has two heated 7-l abrasive powder mixing chambers that can be switched on automatically. The mixers are fed sequentially from a large powder hopper located on a slide above the mixers. The internal surfaces of the powder chambers and the delivery lines and nozzles are diamond lined for high abrasion resistance. The unit is designed to interface with an automated robotic system. A large-diameter nozzle with an integrated vacuum shield provides large surface coverage and high material removal rates. The nozzle can be attached to a robot for fully automatic operation.

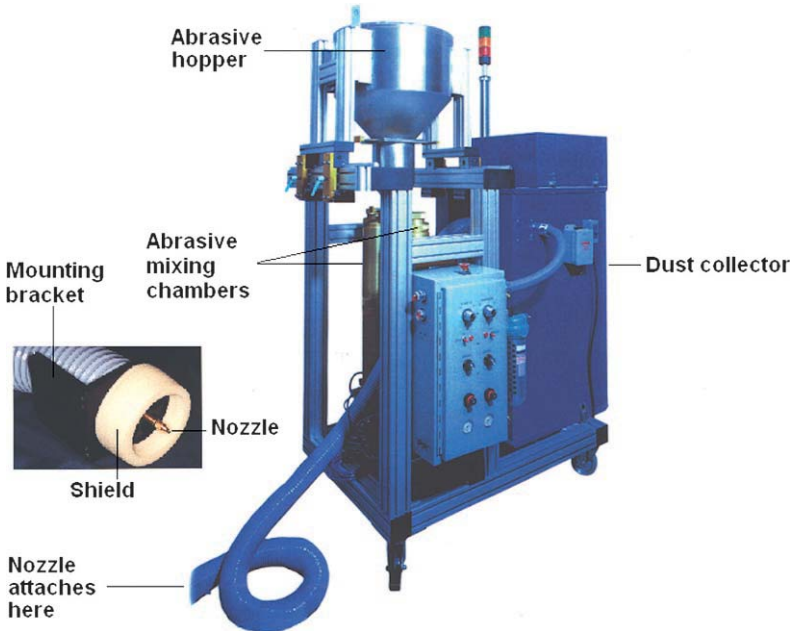


FIGURE 14.4 A production-scale microabrasive processing unit designed for continuous cooperation. The nozzle assembly is shown in the left figure with a dust shield and a bracket to attach to a robotic arm for automatic operation.⁸



FIGURE 14.5 A work station with an integrated microabrasive blasting unit.¹⁸⁴ (Courtesy of Vani-man Manufacturing Company.)

Microabrasive units can be integrated into fixed work stations for performing multiple operations simultaneously. An example of such an integrated system is shown in [Figure 14.5](#).

Multiple microabrasive units can be configured in a multiple nozzle assembly that can be integrated with advanced machining platforms for high-volume, high-throughput processing of parts such as stents, bone screws, pacemakers, and other medical devices that require a high degree of accuracy and repeatability. An example of such a system is shown in [Figure 14.6](#).¹⁷¹ This system is designed to operate in a production environment with minimum operator interaction. It features up to four axes of coordinated abrasive control for precision blasting. The added motion capability increases the range of part shape and size that can be effectively processed on this platform. All aspects of the abrasive blasting process are controlled by the system software. It automatically sets all of the necessary variables including pressure, travel path, and nozzle speed, reducing errors to a minimum.

Safety concerns are important in any blasting operation and microabrasive processing is no exception. If processing is performed in an open and/or confined environment, eye and ear protection should be worn, as well as protective clothing when working on hazardous materials. As with all blasting operations the blast nozzle should never be pointed at anyone or serious injury can occur.

Another consideration is environmental and health concerns. An optimal abrasive is expected to have the following operating characteristics:

- Low levels of dusting under normal usage conditions.
- Low health risk, including low levels of hazardous and toxic contaminants, such as asbestos and arsenic.
- Low environmental impact during storage, use, and disposal.
- Durability of the abrasive to allow collection, reclamation, and reuse.

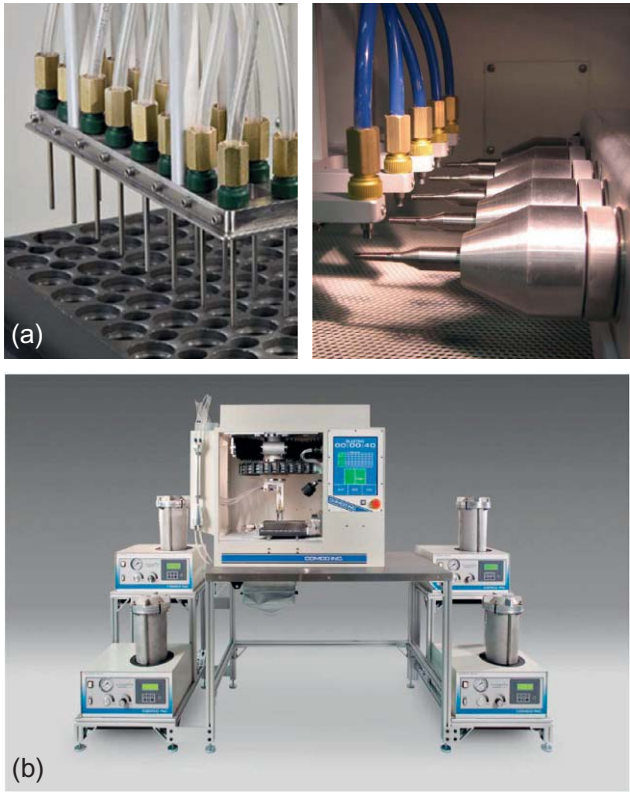


FIGURE 14.6 High-precision, high-volume microabrasive processing of medical parts. The multiple nozzle assemblies (a) are integrated with an automated machining platform (b).¹⁷¹ (Courtesy of Comco, Inc.)

The respiratory effects of the abrasive on the operator are of high concern in a semi-contained non-contained operating environment. Approved respiratory equipment should be provided and monitoring should be performed throughout the job. In confined areas, the toxicity of the abrasive may require the use of additional safety and engineering controls. Additionally, the toxicity of the contaminants being removed often makes disposal of the contaminant/abrasive mixture challenging. The effect of spent abrasive on the soil must also be considered.

14.6 COST CONSIDERATIONS

Small benchtop microabrasive processing systems can cost less than \$500 for simple cleaning applications. More sophisticated industrial systems that are integrated into automated machining platforms with automated nozzle positioning for continuous high-precision operations and high throughput can cost

several hundred thousand dollars. Besides the capital costs, there are operating and material costs for consumables (powders, nozzles, filters) that must be considered. Even so, microabrasive processing is often cost competitive with other conventional cleaning operations. For example, a comparison of the costs associated with removal of polyurethane coating from printed circuit boards (1500 units) showed that microabrasive cleaning resulted in annual savings of nearly \$150,000 compared with traditional chemical methods for cleaning.¹⁸⁸

Recently, a cost optimization model has been developed that considers the effect of many process variables on the overall cleaning costs, including air pressure, nozzle diameter, nozzle wear rate, power costs, consumables costs, cleaning rate, labor rates and costs.¹⁸⁹ This model can be used to optimize the operating conditions of the abrasion process, which can lead to shorter cleaning times and cost savings.

14.7 ADVANTAGES AND DISADVANTAGES

There are many advantages to using microabrasive technology for precision cleaning and processing applications, including versatility, precision, ease-of-use, and low costs. However, there are several disadvantages, such as residual contamination and substrate degradation, which may limit the application of microabrasive technology for certain critical cleaning applications. The advantages and disadvantages are given below.

14.7.1 Advantages

1. The operating parameters can be precisely controlled for processing parts requiring a high degree of accuracy and repeatability, such as medical devices and electronics.
2. The process is sufficiently gentle to be used on delicate surfaces such as fossils, historical, artwork, and eggshells.
3. The process can be employed as a cleaning system to remove surface contaminants such as thick organic films and large inorganic and metal particles and debris.
4. It is very versatile because of the wide range of abrasive powders. The system can remove both metal and hard oxide coatings quickly and effectively.
5. It can be used to fabricate complex parts from refractory metals and ceramics to high tolerances.
6. There are minimal thermal and mechanical stresses on the sample surface.
7. The system is designed for simple operation in manual or automated mode. A high degree of automation is possible.
8. The system is designed to minimize the amount of waste generated. No toxic abrasives are used.

9. This is an environmentally friendly process. A unique vacuum shield and double filtration ensure that no contaminants are released to the environment.
10. It can be used on materials contaminated with radioactive and hazardous substances, and in radioactive environments.
11. Microabrasive units can be scaled with multiple nozzle assemblies for high-volume, high-throughput processing.
12. The system has a high degree of portability. It can be used as a self-contained manual unit for field use or it can be installed as a fixed installation.
13. It has low capital and operating costs.

14.7.2 Disadvantages

1. The substrate of critical components (optical, semiconductor, microelectronic) is likely to be affected by the microabrasive, although the surface is uniformly roughened and it can be controlled to within $\pm 1 \mu\text{m}$.
2. A small amount of the abrasive residue left on the surface could lead to infections in dental applications.
3. There may be contamination risks from abrasive powders liberated in surgical environments such as oral surgery.^{190,191}
4. The process cannot be used in a cleanroom environment.
5. Management of used abrasive can be expensive, including separation and disposal.
6. Abrasive blasting is a line-of-sight process.
7. It requires the use of personnel safety equipment.

14.8 APPLICATIONS

Air abrasion for dentistry was one of the first applications of microabrasive technology in 1945.¹⁹² Since then it has been developed as a minimally invasive dental technique for various dental treatments such as cavity preparation and tooth surface preparation.^{193–211} The versatility of microabrasive technology has been demonstrated by a wide variety of other applications in many different industries.^{169–184,212–269} Many of these applications were discussed previously.⁸ Some of the recent applications are discussed below.

14.8.1 Removal of Coatings

Conformal coatings can be broadly classified as either plastic or resin materials that are applied to electronic devices, medical implants or instrumentation, and automotive products to protect them from various forms of contamination. There are times when these coatings must be removed, either entirely or in specific areas, because of a component failure, to expose a test point, or when older



FIGURE 14.7 Selective removal of coating from a guide wire using sodium bicarbonate abrasive.¹⁷¹ (Courtesy of Comco, Inc.)

devices are returned from the field for repair. Selective removal of a conformal coating can also eliminate the need for costly and time-consuming masking of parts prior to application of the coating. Microabrasive cleaning will safely and selectively remove most types of conformal coatings from various components, including printed circuit boards, without causing mechanical or ESD damage (Figure 14.7). Sodium bicarbonate is an ideal abrasive for coating removal application because of its physical properties and water solubility. The particles are sharp enough to cut through the coating, but soft enough to not damage the part. It is ideal for interventional use because the abrasive can be completely washed off. The coating removal process is fast, environmentally friendly, and cost effective.¹⁸⁸ Also, it does not typically require highly trained personnel.

Microabrasive cleaning is effective in uniformly removing completely the thin ($\sim 2\text{--}3\text{ }\mu\text{m}$) multiple metal (Inconel) and hard oxide (TiO_2 or ZrO_2) coatings on the surface of glass substrates in optoelectronic components. Using $17\text{-}\mu\text{m}$ alumina powder, the coating could be removed from an area of 0.05 m^2 within a few seconds. Inspection of the samples showed the substrate was roughened by the abrasive, although the roughening was uniform over the entire same area and it could be controlled to with $\pm 1\text{ }\mu\text{m}$.⁸ No damage to the substrate was observed in visual or microscopic examination of the surfaces after cleaning.

14.8.2 Etching Microfluidic Devices

Microabrasive blasting can be used for direct etching of microfluidic networks.^{225,226,228,250,263} This method allows creation of features down to $50\text{ }\mu\text{m}$ with a total processing time (design to device) of less than 1 day. The

effectiveness of the process was demonstrated in cationic mode by separating alkali ions (K^+ , Na^+ , and Li^+) and in the anionic mode by separating fumaric, malic, and citric acids.²²⁹

14.8.3 Dental Applications

Microabrasive dentistry is a conservative, less traumatic, alternative to a high-speed drill, which allows doctors to selectively remove decay, leaving healthier tooth structure intact. Since the procedure is less invasive to healthy tooth structure, the microabrasion process can be accomplished in less time and often without the need for anesthesia. It has been established for a wide range of dental applications, including caries treatment, preparation of cavities and prophylaxis, tooth surface preparation for enamel bonding, removal of faulty composite restorations, and removal of superficial stains and tooth discoloration.^{193–211}

Figure 14.8 shows fissure caries on a molar tooth removed with 27- μ m alumina powder with minimal cavity preparation.

14.8.4 Deburring of Machined Components

Microabrasive blasting is effective in deburring and cleaning of small and complex components in the manufacturing process. The process is most effective on soft metals and alloys, such as aluminum and brass, but it can also be used to deburr steel and iron. It has also been successfully applied on non-metallic materials such as polyether ether ketone (PEEK), although the process parameters must be optimized for the application.¹⁷¹ The flowing microabrasive powder stream and the miniature nozzles make it possible to access the tight spaces and corners to deburr a complex part. The system can also be integrated into existing manufacturing and assembly lines to perform the deburring operation automatically.

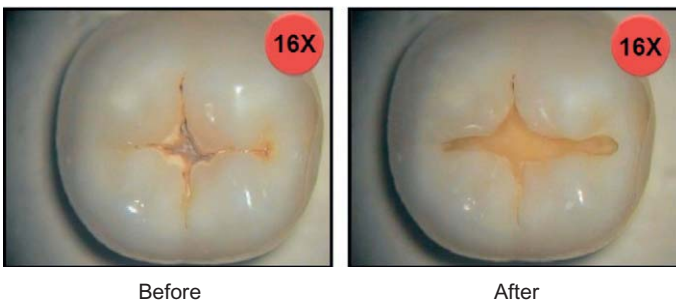


FIGURE 14.8 Microabrasion with 27 μ m alumina powder used to remove fissure caries on a molar tooth.²⁰⁵

14.8.5 Medical Device Processing

Microabrasive blasting technology is increasingly being used by medical device manufacturers worldwide to process the parts that may have unacceptable surface finishes or to clean the tooling used to fabricate the medical devices.^{171,172,184}

- *Guide wires/catheters.* Microabrasive blasting is used to easily remove polymer coatings from small areas on guide wires and catheters (see Figure 14.7). This process is also very efficient at preparing surfaces for bonding and coating adhesion.
- *Implants.* Because implants are usually small and complex, conventional deburring and texturing is very often not possible. Microabrasive blasting provides the precision and accuracy needed and the benefit of being able to select from many types of blasting media which are safe for use in implants (see Figure 14.9). A multi-nozzle assembly can be employed for high-volume, high-throughput processing (see Figure 14.6).
- *Needles/tubes.* When grinding tubes and needles, burrs can form on tubes and needles during grinding and machining of holes and slots. Microabrasive blasting removes these burrs efficiently and effectively. Both metallic and non-metallic components can be treated.
- *Stents.* During the manufacturing process of stents, unwanted material can be present on the surface which needs to be removed. Microabrasive blasting can be used to texture the stent before the application of various coatings without concern for dimensional changes of the stent.
- *Injection molding.* Mold cavity maintenance is critical to medical device manufacturing because the products tend to be smaller and more intricate. The microabrasive process will clean the mold without dimensional changes.
- *Electronics.* Electronic devices can be effectively and easily treated with microabrasive processing to remove conformal coatings, excess encapsulation materials, and other activities.

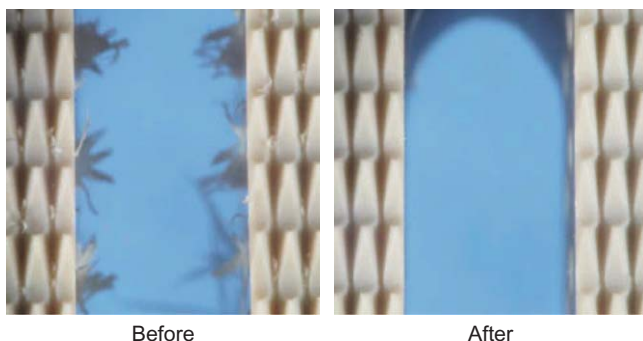


FIGURE 14.9 Machined PEEK implant with burrs (left) that were removed after microabrasive blasting with sodium bicarbonate powder.¹⁷¹ (Courtesy of Comco, Inc.)

14.8.6 Fabrication of Complex Three-Dimensional Structures

Brittle materials like glass or silicon are also amenable to fabrication of high-precision three-dimensional microstructures.^{219–224,229,232,234,238–242,267} The process is characterized by erosion rates up to several hundred micrometers per minute using 30- μm alumina particles. By employing boron nitride or diamond powders with a special multi-nozzle assembly, the microabrasive system has been used to precisely fabricate burr-free complex parts directly from 100- μm thick Mo or Mo-Re foil.⁸ The production rate for fabricating these parts with the microabrasive system is comparable to an Nd-YAG laser machining system; however, the cost of the microabrasive system is less than half the cost of the laser machining system (\$500,000). In addition, laser machining leaves burrs on the surface of the parts, requiring further acid chemical cleaning to produce the final finish. Waste treatment and disposal of toxic and corrosive acids significantly increase the total cost of laser processing and contribute to environmental pollution.

14.8.7 Cutting Brittle Materials

Microabrasive is an effective process for fabricating micro-grooves in brittle materials such as quartz and glass.^{20,256} Groove depth increases with an increase in air pressure and nozzle impact angle and a decrease in nozzle traverse speed. By contrast, a narrow groove can be achieved by decreasing air pressure and increasing jet impact angle and nozzle traverse speed. Further, minimum kerf taper can be achieved by increasing air pressure and abrasive mass flow rate and decreasing the nozzle traverse speed with the orthogonal impact angle, while a better surface finish can be produced with a lower air pressure and jet impact angle and a higher nozzle traverse speed. For solar panel applications grooves have been precisely cut through hard and soft coatings of 0.5–10 μm thickness on glass panels with an automated microabrasive unit at fast rates of up to 40–50 cm/s. The grooves were 50–75 μm wide through each coating layer, and there was no damage to the underlying surface. The estimated capital and operating costs of an automated integrated system for this specific application are nearly 20–30% less than that for comparable laser cutting systems. An alternative method for cutting grooves is a high-precision mechanical diamond stylus cutting system. The total cost of this mechanical system is comparable to the microabrasive system; however, the diamond stylus system cannot provide the necessary degree of precision and control.

14.8.8 Miscellaneous Applications

Microabrasive technology has been successfully employed in many other applications including: cleaning of historical granite in historically significant structures^{179,236,269}; fossil preparation²⁵⁷; cleaning hardened coatings off the surface of containers that are used or will be reused in processing operations²²⁵; trimming of a capacitor in quartz-crystal oscillators for wrist-watch and/or clock

assemblies²¹³; surface bombardment with microabrasive to modify the surface of a metal such as nitinol for use as a drug-eluting stent^{259,260}; polishing of optical fiber end faces to achieve high surface roughness²³⁴; fabrication of bulk ceramic lead zirconate titanate (PZT) transducers²⁶²; micromachining of polytetrafluoroethylene (PTFE), polydimethylsiloxane (PDMS), and carbon steel at cryogenic temperatures²⁶⁴; micromachining of acrylic and polycarbonate polymers^{247,248}; surface microblasting for improving the performance and life of coated cutting tools and tips.^{238,244,245,249,253}

Other diverse and uncommon applications for which microabrasive technology has been considered include: spraying pulverized material in agricultural applications and even as spray for personal defense²¹²; cleaning of surgical and medical instruments prior to sterilization; recovery of precious metals from used electronic components; precise selective decontamination of parts contaminated with hazardous or radioactive materials; precision cutting and processing of explosives and munitions; precision sampling from the surface of radioactively contaminated lead components for chemical analysis; disassembly of nuclear weapons; texturing of patterns on athletic shoe uppers.

14.9 SUMMARY AND CONCLUSIONS

Microabrasive technology is a highly versatile, automated and self-contained, non-solvent precision cleaning and processing system. This established process uses a precisely controlled stream of micro-fine particles at high velocity to perform operations such as surface cleaning, cutting, and micromachining of fragile, brittle, and other materials. The basic mechanisms of particle impact can be applied to surface film and particle removal. Several applications have been described.

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